



# Bhabha University, Bhopal (M.P.)

## Scheme of Examination

**Program: Bachelor of Technology**  
**Semester : III**

**Branch: Information Technology**  
**Specialization: Information Technology**

**Session: 2020-21**

| S..No. | Subject Code | Subject Name                                     | Maximum Marks Allotted |                |                  |           |                      | Total Marks | Contact Hours per week |   |    | Total Credits |
|--------|--------------|--------------------------------------------------|------------------------|----------------|------------------|-----------|----------------------|-------------|------------------------|---|----|---------------|
|        |              |                                                  | Theory                 |                |                  | Practical |                      |             |                        |   |    |               |
|        |              |                                                  | End Sem.               | Mid Sem. Exam. | Quiz/ Assignment | End Sem   | Term work            |             |                        |   |    |               |
|        |              |                                                  |                        |                |                  |           | Lab Work & Sessional |             |                        |   |    |               |
| 1.     | CH-31        | Energy & Environmental Engineering               | 70                     | 20             | 10               | -         | -                    | 100         | 3                      | 1 | -  | 4             |
| 2.     | MA-31        | Mathematics III                                  | 70                     | 20             | 10               | -         | -                    | 100         | 3                      | 1 | -  | 4             |
| 3.     | IT-31        | Data Structure                                   | 70                     | 20             | 10               | 30        | 20                   | 150         | 3                      | - | 2  | 4             |
| 4.     | IT-32        | Object Oriented Programming & Methodology (OOPM) | 70                     | 20             | 10               | 30        | 20                   | 150         | 3                      | - | 2  | 4             |
| 5.     | IT-33        | Digital Circuits & System(DCS)                   | 70                     | 20             | 10               | 30        | 20                   | 150         | 3                      | - | 2  | 4             |
| 6.     | IT-34        | Computer Programming Python                      | -                      | -              | -                | 30        | 20                   | 50          | -                      | - | 4  | 2             |
| 7.     | IT-35        | Group Discussion (GD)/SEMINAR                    | -                      | -              | -                | -         | 50                   | 50          |                        |   | 4  | 2             |
|        |              | TOTAL                                            | 350                    | 100            | 50               | 120       | 130                  | 750         | 15                     | 2 | 14 | 24            |

\*End Semester Examination will be of three hours.

**PRACTICALS SUBJECT** IT- 31: Datastructure  
IT-33: DCS

IT-32: OOPM  
IT-34: Computer programming Python.

IT-35: GD/Seminar

|              |               |                |
|--------------|---------------|----------------|
| 1 Hr Lecture | 1 Hr Tutorial | 2 Hr Practical |
| 1 Credit     | 1 Credit      | 1 Credit       |

BHABHA UNIVERSITY BHOPAL



# **BHABHA UNIVERSITY, BHOPAL**

## **COURSE CURRICULUM FACULTY OF ENGINEERING**

### **Undergraduate Programme**

**(Courses effective from Academic Year 2020-21)**

### **SYLLABUS OF COURSES TO BE OFFERED “Bachelor of Technology”**

**Disclaimer:** The undergraduate programme in Bachelor of Technology syllabus & scheme is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

**Dean faculty of Engineering, Bhabha University**

**NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026**

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# **BACHELOR OF TECHNOLOGY**

## **III SEMESTER**

### **INFORMATION TECHNOLOGY(IT)**

**NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026**

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## **BHABHA UNIVERSITY , BHOPAL**

### **INFORMATION TECHNOLOGY III SEMESTER**

#### **CH-31 Energy & Environmental Engineering**

##### **UNIT 1: Introduction to Energy Science**

Introduction to energy systems and resources; Introduction to Energy, sustainability & the environment; Overview of energy systems, sources, transformations, efficiency, and storage; Fossil fuels (coal, oil, oil-bearing shale and sands, coal gasification) - past, present & future, Remedies & alternatives for fossil fuels - biomass, wind, solar, nuclear, wave, tidal and hydrogen; Sustainability and environmental trade-offs of different energy systems; possibilities for energy storage or regeneration (Ex. Pumped storage hydro power projects, superconductor-based energy storages, high efficiency batteries)

##### **UNIT 2: Ecosystems**

Concept of an ecosystem; Structure and function of an ecosystem; Producers, consumers and decomposers; Energy flow in the ecosystem; Ecological succession; Food chains, food webs and ecological pyramids; Introduction, types, characteristic features, structure and function of the following ecosystem (a) Forest ecosystem (b) Grassland ecosystem (c) Desert ecosystem (d) Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

##### **UNIT 3: Biodiversity and its conservation**

Introduction – Definition: genetic, species and ecosystem diversity; Bio-geographical classification of India; Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values; Biodiversity at global, National and local levels; India as a mega-diversity nation; Hot-spots of biodiversity; Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; Endangered and endemic species of India; Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

##### **UNIT 4: Environmental Pollution**

Definition, Cause, effects and control measures of Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution, Nuclear hazards; Solid waste Management: Causes, effects and control measures of urban and industrial wastes; Role of an individual in prevention of pollution; Pollution case studies; Disaster management: floods, earthquake, cyclone and landslides.

##### **UNIT 5: Social Issues and the Environment**

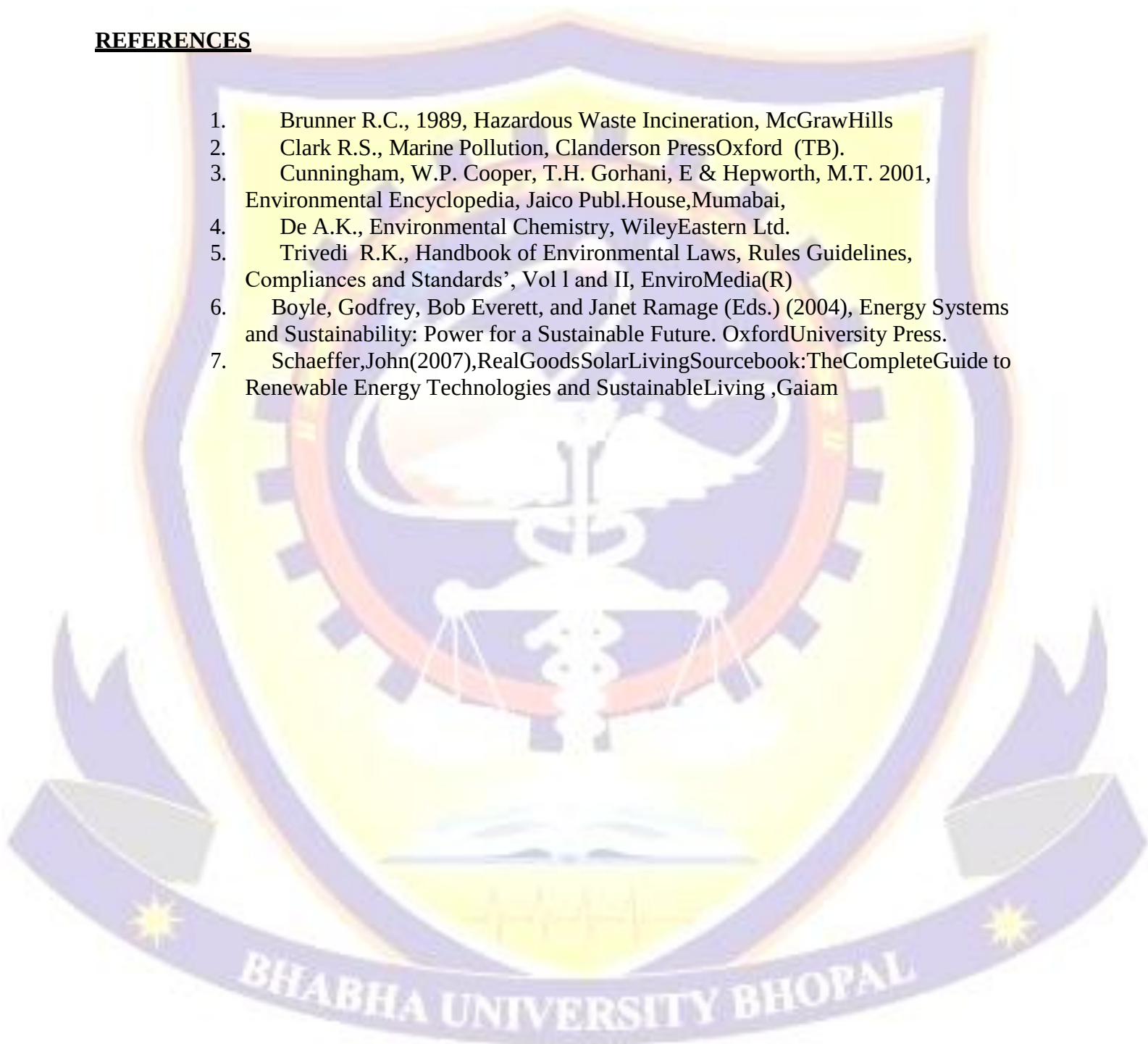
From Unsustainable to Sustainable development; Urban problems related to energy; Water conservation, rain water harvesting, watershed management; Resettlement and rehabilitation of people; its problems and concerns. Case Studies  
Environmental ethics: Issues and possible solutions. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies  
Wasteland reclamation; Consumerism and waste products; Environment Protection Act; Air (Prevention and Control of Pollution) Act; Water (Prevention and control of Pollution) Act; Wildlife Protection Act; Forest Conservation Act; Issues involved in enforcement of environmental legislation; Public awareness.



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**INFORMATION TECHNOLOGY III SEMESTER**

**REFERENCES**

1. Brunner R.C., 1989, Hazardous Waste Incineration, McGrawHills
2. Clark R.S., Marine Pollution, Clanderson PressOxford (TB).
3. Cunningham, W.P. Cooper, T.H. Gorhani, E & Hepworth, M.T. 2001, Environmental Encyclopedia, Jaico Publ.House,Mumabai,
4. De A.K., Environmental Chemistry, WileyEastern Ltd.
5. Trivedi R.K., Handbook of Environmental Laws, Rules Guidelines, Compliances and Standards', Vol I and II, EnviroMedia(R)
6. Boyle, Godfrey, Bob Everett, and Janet Ramage (Eds.) (2004), Energy Systems and Sustainability: Power for a Sustainable Future. OxfordUniversity Press.
7. Schaeffer,John(2007),RealGoodsSolarLivingSourcebook:TheCompleteGuide to Renewable Energy Technologies and SustainableLiving ,Gaia





## **BHABHA UNIVERSITY , BHOPAL**

### **INFORMATION TECHNOLOGY III SEMESTER**

#### **MA-31Mathematics III**

##### **UNIT 1: Numerical Methods – 1**

Solution of polynomial and transcendental equations – Bisection method, Newton-Raphson method and Regula-Falsi method. Finite differences, Relation between operators, Interpolation using Newton's forward and backward difference formulae. Interpolation with unequal intervals: Newton's divided difference and Lagrange's formulae.

##### **UNIT 2: Numerical Methods – 2**

Numerical Differentiation, Numerical integration: Trapezoidal rule and Simpson's 1/3rd and 3/8 rules. Solution of Simultaneous Linear Algebraic Equations by Gauss's Elimination, Gauss's Jordan, Crout's methods, Jacobi's, Gauss-Seidal, and Relaxation method.,

##### **UNIT3: Numerical Methods – 3**

Ordinary differential equations: Taylor's series, Euler and modified Euler's methods. Runge-Kuttamethodoffourthorderforsolvingfirstandsecondorderequations.Milne's andAdam'spredicator-correctormethods.Partialdifferentialequations:Finitedifference solution two dimensional Laplace equation and Poisson equation, Implicit and explicit methods for one dimensional heat equation (Bender- Schmidt and Crank-Nicholson methods), Finite difference explicit method forwaveequation.

##### **UNIT 4: Transform Calculus**

Laplace Transform, Properties of Laplace Transform, Laplace transform of periodic functions.FindinginverseLaplacetransformbydifferentmethods,convolutiontheorem. Evaluation of integrals by Laplace transform, solving ODEs by Laplace Transform method,Fouriertransforms.

##### **UNIT 5: Concept of Probability**

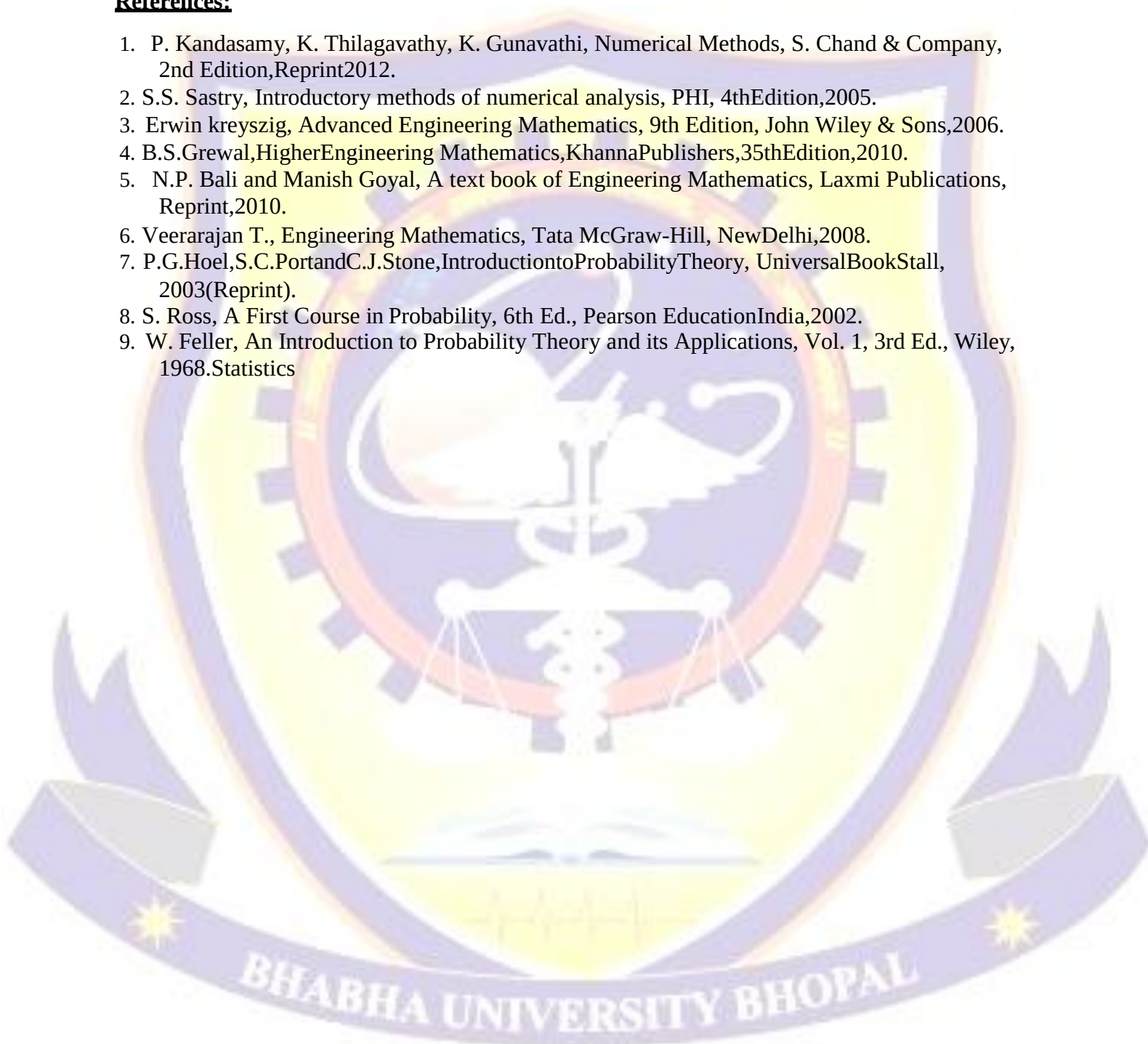
Probability Mass function, Probability Density Function, Discrete Distribution: Binomial, Poisson's, Continuous Distribution: Normal Distribution, ExponentialDistribution.



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**References:**

1. P. Kandasamy, K. Thilagavathy, K. Gunavathi, Numerical Methods, S. Chand & Company, 2nd Edition, Reprint 2012.
2. S.S. Sastry, Introductory methods of numerical analysis, PHI, 4th Edition, 2005.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
4. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 35th Edition, 2010.
5. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2010.
6. Veerarajan T., Engineering Mathematics, Tata McGraw-Hill, New Delhi, 2008.
7. P.G. Hoel, S.C. Portant C.J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003 (Reprint).
8. S. Ross, A First Course in Probability, 6th Ed., Pearson Education India, 2002.
9. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968. Statistics





## **BHABHA UNIVERSITY , BHOPAL**

### **INFORMATION TECHNOLOGY III SEMESTER**

#### **IT-31 DataStructure**

**Unit 1:** Introduction Data, data type, data object. Types of data structure – primitive & non-primitive , linear & non-linear. Operations on data structures – traversing, searching , inserting , deleting. Complexity analysis – worst case, best case, average case. Time – space trade off , algorithm efficiency, asymptotic notations – big oh , omega ,theta.

**Unit 2:** Arrays & Structure Introduction , declaration of arrays , operations on arrays – inserting , deleting , merging of two arrays , 1 dimensional & 2 dimensional arrays, row & column major representation , address calculation in array , storing values in arrays , evaluation of polynomial – addition & representation. Searching & sorting – Introduction , sequential search, binary search , Fibonacci search , indexed sequential search, hashed search. Types of sorting with general concepts–bubble,heap,insertion,selection,quick,heap,shell,bucket,radix and mergesort.

**Unit 3:** Stacks & Queues Basic concept of stacks & queues, array representation of stacks, operation on stacks–push,pop,create,getTop,empty,linked representation of stack,multiple stack. Application of stack – Conversion: infix , prefix , postfix and evaluation of arithmetic expression. Linked representation of queue, operations on queue – insertion & deletion. Types of queue with functions – circular , deque , priority queue. Applications of queues – job scheduling, Josephus problem.

**Unit 4:** Linked List Introduction – basic terminology , memory allocation & deallocation for linked list. Linked list variants – head pointer , head node , types linked list – linear & circular linked list. Doubly linked list , creation of doubly list, deletion of node from doubly linked list, insertion of a node from doubly linked list, traversal of doubly linked list. Circular linked list – singly circular linked list , circular linked list with header node , doubly circular linked list. Applications of linked list – polynomial representation & garbage collection.

**Unit 5:** Trees Basic terminology – general tree , representation of general tree, types of trees, binary tree- realization and properties , traversal in binary trees – inorder , preorder , postorder , applications of trees. Graph- Basic Terminologies and representations, Graph search and traversal algorithms.

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**List of Experiments**

1. Write a program to search an element in the array using Linear and Binary Search.
2. Write a program to perform the following operation in Matrix:
  1. Addition 2. Subtraction 3. Multiplication 4. Transpose
3. Write a program to perform the following operation on strings using string functions:
  1. Addition 2. Copying 3. Reverse 4. Length of String
4. Write program for implementing the following sorting methods to arrange a list of integers in ascending order:
  - a) Quick sort b) Selection sort c) Insertion sort d) Merge sort
5. Write a program that uses stack operations to convert a given infix expression into its postfix equivalent.
6. Write a program to merge two sorted array into one sorted array.
7. Write a program to implement stack using array and linked list.
8. Write a program to implement queue and circular queue using array.
9. Write a program to insert an element in the beginning and end of singly linked list.
10. Write a program to insert an element at any position in singly and doubly linked list.
11. Insert and delete a node at any position in doubly linked list.
12. Write a program of Tower of Hanoi.
13. Write a program that uses functions to perform the following:
  - a) Create a binary search tree of integers.
  - b) Traverse the above Binary search tree non recursively in inorder.

**References :**

- 1 Varsha H. Patil "Data Structure Using C++" Oxford.
- 2 Rajesh K. Shukla "Data Structures Using C & C++" Wiley India.
- 3 Reema Thareja "Data Structure Using C" Oxford.
- 4 D. S Malik "Data Structure Using C++ " Second Edition Cengage.
- 5 Kushwaha and Mishra "Data Structure: A programming Approach with C", PHI Learning.
- 6 A. K Sharma "Data Structure Using C" Pearson.
- 7 Ellis Horowitz, Sartaj Sahni, "Fundamentals of Data Structures", Computer Science Press

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## **BHABHA UNIVERSITY , BHOPAL**

### **INFORMATION TECHNOLOGY III SEMESTER**

#### **IT-32 Object Oriented Programming & Methodology**

##### **Unit I- Introduction:**

Object oriented programming, Introduction, Application, characteristics, difference between object oriented and procedure programming, Comparison of C and C++, Cout, Cin, Data Type, Type Conversion, Control Statement, Loops, Arrays and string arrays fundamentals, Function, Returning values from functions, Reference arguments, Overloaded function, Inline function, Default arguments, Returning by reference.

##### **Unit II- Object and Classes:**

Implementation of class and object in C++, access modifiers, object as data type, constructor, destructor, Object as function arguments, default copy constructor, parameterized constructor, returning object from function, Structures and classes, Classes objects and memory, static class data, Arrays of object, Arrays as class Member Data, The standard C++ String class, Run time and Compile time polymorphism.

##### **Unit III- Operator overloading and Inheritance:**

Overloading unary operators, Overloading binary operators, data conversion, pitfalls of operators overloading, Concept of inheritance, Derived class and base class, access modifiers, types of inheritance, Derived class constructors, member function, public and private inheritance.

##### **Unit IV- Pointer and Virtual Function:**

Addresses and pointers, the address-of operator & pointer and arrays, Pointer and Function pointer, Memory management: New and Delete, pointer to objects, debugging pointers, Virtual Function, friend function, Static function, friend class, Assignment and copy initialization, this pointer, dynamic type information.

##### **Unit V- Streams and Files:**

Streams classes, Stream Errors, Disk File I/O with streams, file pointers, error handling in file I/O with member function, overloading the extraction and insertion operators, memory as a stream object, command line arguments, printer output, Function templates, Class templates Exceptions, Containers, exception handling.

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**List of Experiments:**

1. Write a program to find out the largest number using function.
2. Write a program to find the area of circle, rectangle and triangle using function overloading.
3. Write a program to implement complex numbers using operator overloading and type conversion.
4. Write a program using class and object to print bio-data of the students.
5. Write a program which defines a class with constructor and destructor which will count number of objects created and destroyed.
6. Write a program to add two private data members using friend function.
7. Write a program using dynamic memory allocation to perform 2x2 matrix addition and subtraction.
8. Write a program to create a stack using virtual function.
9. Write a program that stores five student records in a file.
10. Write a program to get IP address of the system.

**References:**

1. E. Balaguruswami, "Object Oriented Programming in C++", TMH.
2. Robert Lafore, "Object Oriented Programming in C++", Pearson.
3. M.T. Somashekare, D.S. Guru, "Object-Oriented Programming with C++", PHI.
4. Herbert Schildt, "The Complete Reference C++", Tata McGraw Hill publication.

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### **INFORMATION TECHNOLOGY III SEMESTER**

#### **IT-33 Digital Circuits and Systems**

##### **Unit I- Number systems and logic gates:**

Decimal, Binary, Octal, Hexadecimal numbers systems and radix conversion. Codes-BCD, excess 3, gray, ASCII. Boolean algebra- Theorems and properties, Boolean functions, canonical and standard forms, De Morgans theorem, digital logic gates, Karnaugh maps.

##### **Unit II- Combinational circuits:**

Introduction to combinational circuits, multilevel NAND, NOR implementation. Designing binary Adders and Subtractors. Decoder, Encoder, Multiplexer, Demultiplexer circuits.

##### **Unit III- Sequential circuits:**

Introduction to Sequential circuits, flip-flops, RS, D, T, JK, M/S JK-flipflops, truth tables, excitation tables and characteristic equations, clocked and edge triggered flipflops, Registers- Definition, serial, parallel, shift left/right registers, Johnson counter, asynchronous and synchronous counters.

##### **Unit IV- Digital logic families:**

Bipolar and unipolar logic families, Digital IC specifications, RTL, DTL, All types of TTL circuits, ECL, IIL, PMOS, NMOS & CMOS Logic.

##### **Unit V- Clocks and timing circuits:**

Bistable, Monostable & Astable multivibrator, Schmitt trigger circuit, Introduction of Analog to Digital & Digital to Analog converters, Display devices, 7 and 16 segment LED display, LCD.

##### **List of Experiments:**

1. Study and verify the operation of AND, OR, NOT, NOR and NAND logic gates.
2. Design all basic logic gates using NOR universal gate.
3. Design all basic logic gates using NAND universal gate.
4. Verification of Demorgan's theorem.
5. Construction and verification of half adder and full adder circuits.
6. Construction and verification of half subtractor and full subtractor circuits.
7. Design of Binary to Grey & Grey to Binary code Converters.
8. Design of BCD to excess-3 code converter.
9. Design and verification of Multiplexer circuit

##### **References:**

1. M. Morris Mano, "Digital logic design", Pearson Education Pvt. Ltd.
2. A Anand Kumar, "Fundamentals of digital circuits", PHI Learning Pvt. Ltd.
3. A K Maini, "Digital Electronics Principles and Integrated Circuits, Wiley India Pvt. Ltd.
4. R P Jain, "Modern Digital Electronics", Tata McGraw-Hill publishing company Ltd.
5. D P Kothari and J S Dhillon, "Digital Circuits and Design", Pearson Education Pvt. Ltd.



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**IT-34Computer Programming Python**

**UNIT 1- Introduction:**

Basic syntax, Literal Constants, Numbers, Variable and Basic data types, String, Escape Sequences, Operators and Expressions, Evaluation Order, Indentation, Input Output, Functions, Comments.

**UNIT 2- Data Structure:**

List, Tuples, Dictionary and Sets.

**UNIT 3-Control Flow:**

ConditionalStatements-If,If-else,NestedIf-else.IterativeStatement-For,While,NestedLoops.  
Control statements - Break, Continue,Pass.

**UNIT 4-Object oriented programming:**

Class and Object, Attributes, Methods, Scopes and Namespaces, Inheritance,Overloading, Hidingandoverriding .

**UNIT 5-Exception:**

Exception Handling, Except clause, Try finally clause, User Defined Exceptions.

**List of Experiments**

- 1) Python Program to find the area of triangle
- 2) Python program to swap two variables
- 3) Program to print Even Numbers up to N
- 4) Program to check whether the given no. is Prime or not (Method I)
- 5) Write a program to create dataframe for 3 student including name and roll numbers. and add new columns for 5 subjects and 1 column to calculate percentage. It should include random numbers in marks of all subjects
- 6) Recursive Function to Print Multiplication Table
- 7) Recursive Function to print Fibonacci series upto N terms.
- 8) Write a Python program to remove duplicates from a list.

**REFERENCES**

- Timothy A. Budd: Exploring python, McGraw-HillEducation.
- R.Nageshwar Rao ,”Python Programming” ,Wiley India ThinkPython:
- Allen B. Downey, O'Reilly Media,Inc
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## **BHABHA UNIVERSITY , BHOPAL**

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#### **IT-35 Group Discussion (GD)/Seminar**

In Seminar or Group Discussion (GD) students have to select general topics related to current affairs or related to subject & make a PPT of the selected topic and present in front of the faculties & students. By doing this, in front of faculties & students, students communication skill will get developed & will be a good presenter. By doing this repetitively students subject knowledge will also get enhanced. Rather than making PPT, student can also discuss in the group of students in front of faculties that is called as Group Discussion.

